

STT 465
Bayesian Data Analysis
Lecture 02: Probability Review I

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Learning Goals

- A working understanding of probability theory
 - Probability
 - Bayes' rule
 - Generating random samples in R

Sports Example: Free Throws

- Suppose a basketball player makes a free throw with probability $p = 0.75$.
- Let $X_i = 1$ if the i th free throw is made and $X_i = 0$ otherwise.
- Then

$$X_i \sim \text{Bernoulli}(0.75).$$

- The number of made free throws in 10 attempts is

$$Y = \sum_{i=1}^{10} X_i \sim \text{Binomial}(10, 0.75).$$

Sports Example: What Is the Probability?

Suppose a player has a 75% free-throw percentage.

What is the probability that the player makes exactly 8 of 10 free throws?

Using the binomial model,

$$P(Y = 8) = \binom{10}{8} (0.75)^8 (0.25)^2.$$

The same probability can be calculated in R:

```
dbinom(8, size = 10, prob = 0.75)
```

Sports Example: Baseball Batting

Suppose a player has probability p of getting a hit in an at-bat.

If we observe n independent at-bats, then

$$Y \mid p \sim \text{Binomial}(n, p).$$

The observed batting average is

$$\hat{p} = \frac{Y}{n}.$$

For example, 30 hits in 100 at-bats gives

$$\hat{p} = 0.30.$$

But how certain are we that the player's true hitting probability is close to 0.30?

Defining Probability on Sets $(\Omega, \mathcal{F}, P)$

- Random experiment examples:
 - An experiment of tossing a die



- An experiment of flipping two coins



Defining Probability on Sets $(\Omega, \mathcal{F}, P)$

- Sample space Ω
 - The set of all the outcomes of a random experiment
 - Each outcome $\omega \in \Omega$ can be thought of as a complete description of a state at the end of the experiment
- Examples
 - An experiment of tossing a die

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

- An experiment of flipping two coins

$$\Omega = \{TT, TH, HT, HH\}$$

Defining Probability on Sets $(\Omega, \mathcal{F}, P)$

- Set of events (or event space) $\mathcal{F}$
 - A set whose elements (i.e. events) $A \in \mathcal{F}$ are subsets of Ω
 - Each element $A \in \mathcal{F}$ can be thought of as a question that you can ask at the end of the experiment
 - Examples
 - An experiment of tossing a die

$$\mathcal{F} = \{\emptyset, \{1\}, \dots, \{6\}, \{1, 2\}, \dots, \{1, 3, 5\}, \dots, \{2, 4, 6\}, \dots, \Omega\}$$

- An experiment of flipping two coins

$$\mathcal{F} = \{\emptyset, \{TT\}, \dots, \{HH\}, \{TT, TH\}, \dots, \{TH, HT\}, \dots, \Omega\}$$

- If there are K elements in Ω , then there are 2^K elements in $\mathcal{F}$

Defining Probability on Sets $(\Omega, \mathcal{F}, P)$

- Set of events (or event space) $\mathcal{F}$
 - $\mathcal{F}$ should satisfy:
 - $\emptyset \in \mathcal{F}$
 - $A \in \mathcal{F} \Rightarrow \Omega \setminus A \in \mathcal{F}$
 - $A_1, A_2, \dots \in \mathcal{F} \Rightarrow \cup_i A_i \in \mathcal{F}$

Defining Probability on Sets $(\Omega, \mathcal{F}, P)$

- Probability measure: A function $P : \mathcal{F} \rightarrow \mathbb{R}$
 - P should satisfy:
 - Non-negativity: $P(A) \geq 0$, for all $A \in \mathcal{F}$
 - Total probability: $P(\Omega) = 1$
 - Countable additivity: If $A_1, A_2, \dots$ are disjoint events (i.e., $A_i \cap A_j = \emptyset$ whenever $i \neq j$), then $P(\cup_i A_i) = \sum_i P(A_i)$
 - These three properties are called **the Axioms of Probability**
 - Examples
 - An experiment of tossing a die

$$P(A_1) = \dots = P(A_6) = 1/6 \Rightarrow P(A_1 \cup A_3 \cup A_5) = \sum_{i \in \{1,3,5\}} P(A_i) = 1/2$$

- An experiment of flipping two coins

$$P(A_{TT}) = \dots = P(A_{HH}) = 1/4 \Rightarrow P(A_{H\cdot}) = P(A_{HT}) + P(A_{HH})$$

Probability Measure P

- Properties of P (suppose A and B are two events $\in \mathcal{F}$)
 - If $A \subseteq B \Rightarrow P(A) \leq P(B)$
 - $P(A \cap B) \leq \min(P(A), P(B))$
 - $P(A \cup B) \leq P(A) + P(B)$
 - $P(\Omega \setminus A) = 1 - P(A)$
 - If $A_1, \dots, A_k$ are a set of disjoint events such that $\bigcup_{i=1}^k A_i = \Omega$, then $\sum_{i=1}^k P(A_i) = 1$

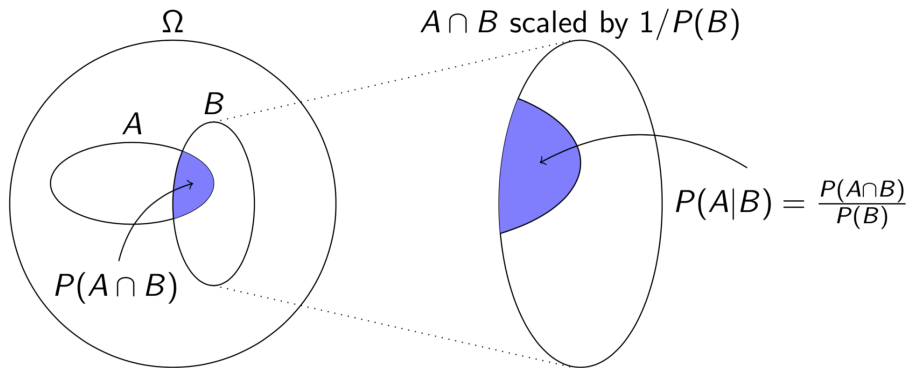
Conditional Probability

- Let B be an event with non-zero probability, then the conditional probability of any event A given event B is defined as

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

- $P(A|B)$ is the probability measure of the event A after observing the occurrence of event B
- Conditional probability satisfies the Axioms of Probability
 - $B \subseteq \Omega$ becomes the new sample space
 - $A \cap B \subseteq B$ becomes the new collection of possible outcomes

Conditional Probability



Sports Example: Conditional Probability

Suppose a football team scores a touchdown on 70% of its red-zone possessions.

Let

$$A = \{\text{team scores a touchdown}\}$$

and

$$B = \{\text{team reaches the red zone}\}.$$

The relevant probability is

$$P(A \mid B).$$

Notice the difference between

$$P(A \mid B) \quad \text{and} \quad P(B \mid A).$$

Conditioning changes the question.

Sports Example: Home Advantage

Suppose we are interested in whether a team wins a game.

Let

$$W = \{\text{team wins}\}$$

and

$$H = \{\text{team plays at home}\}.$$

We may want to compare

$$P(W \mid H)$$

with

$$P(W \mid H^c).$$

This is a simple introduction to conditional probability in sports data.

Sports Example: Correlation Is Not Independence

Consider a basketball team.

Let

X = number of three-point attempts

and

Y = number of three-point makes.

A natural relationship is that more attempts tend to produce more makes.

Thus X and Y are not independent.

This illustrates why dependence between variables matters when building statistical models for sports data.

In-class Practice: Conditional Probability

- In an experiment of tossing a die, what is the probability of getting an odd number, given that the observed number is less than four?

- Let

$$A = \{1, 3, 5\}, \quad B = \{1, 2, 3\}.$$

- Find $P(A \mid B)$.

Solution: Conditional Probability

- $A \cap B = \{1, 3\}$, so

$$P(A \cap B) = \frac{2}{6} = \frac{1}{3}, \quad P(B) = \frac{3}{6} = \frac{1}{2}.$$

- Therefore,

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{1/3}{1/2} = \boxed{\frac{2}{3}}.$$

Independence

- Independence is equivalent to say that observing event B does not have any effect on the probability of observing event A

$$P(A) = P(A|B) = \frac{P(A \cap B)}{P(B)}$$

- Two events are independent if and only if

$$P(A \cap B) = P(A)P(B)$$

or equivalently, $P(A|B) = P(A)$ if B is an event with non-zero probability, i.e. $P(B) > 0$

In-class Practice: Independence

- In an experiment of tossing a die, are the events

$$A = \{1, 3, 5\}, \quad B = \{1, 2, 3\}$$

independent?

- Check whether

$$P(A \cap B) = P(A)P(B).$$

Solution: Independence

- We have

$$P(A) = \frac{1}{2}, \quad P(B) = \frac{1}{2}, \quad P(A \cap B) = \frac{1}{3}.$$

- But

$$P(A)P(B) = \frac{1}{4}.$$

- Since $\frac{1}{3} \neq \frac{1}{4}$,

A and B are not independent.

Bayes' Theorem

- Derived by writing the joint probability of both A and B in two ways:

$$P(A \cap B) = P(A|B)P(B) = P(B|A)P(A)$$

- Suppose B is an event with non-zero probability, then the statement of Bayes' Theorem is:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

- $P(A|B)$: A conditional probability that indicates the probability of event A occurring given that B occurs
- $P(B|A)$: A conditional probability that indicates the probability of event B occurring given that A occurs
- $P(A)$ and $P(B)$: probabilities of observing A and B independently of each other

Sports Example: Bayes' Rule

Suppose a sports analyst classifies teams into two types:

- S : a strong team,
- W : a weak team.

Suppose a team wins its first three games.

We may ask

$$P(S \mid \text{three wins}).$$

Bayes' rule gives

$$P(S \mid \text{three wins}) = \frac{P(\text{three wins} \mid S)P(S)}{P(\text{three wins})}.$$

Past beliefs and new evidence are combined.

Sports Example: Prior Matters

Suppose two analysts observe the same three-game winning streak.

Analyst 1 initially believes

$$P(S) = 0.2.$$

Analyst 2 initially believes

$$P(S) = 0.7.$$

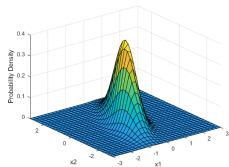
Even with the same likelihood model,

$$P(S \mid \text{three wins})$$

can be different because the prior probabilities are different.

This is the basic Bayesian idea:

prior belief + new evidence $\longrightarrow$ updated belief.

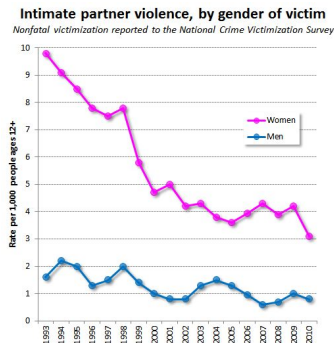


- Orenthal J. Simpson: A running back
 - 40-th greatest player in the NFL list
- Arrested and charged with the murders of his ex-wife in 1994
 - Released in 2017
- His attorney defended: “Only 0.1% of the men who abuse their wives end up murdering them. The fact that Simpson abused his wife is irrelevant to the case”
- Ignore the fact that Simpson’s wife was actually murdered

- A = husband abuse wife
- B = wife is murdered
- $P(B|A) = P(\text{wife is murdered}|\text{husband abuse wife}) = 0.001$
- $P(A|B) = P(\text{husband abuse wife}|\text{wife is murdered}) = ?$

$$\begin{aligned}P(A|B) &= \frac{P(B|A)P(A)}{P(B)} \\&= 0.001 \frac{P(A)}{P(B)} = 0.001 \times \frac{\text{husband abuse wife}}{\text{wife is murdered}} \\&= 0.001 \times 990 \\&= 0.99\end{aligned}$$

- The probability of husband abuse wife $P(A) = 9/1,000 = 0.009$

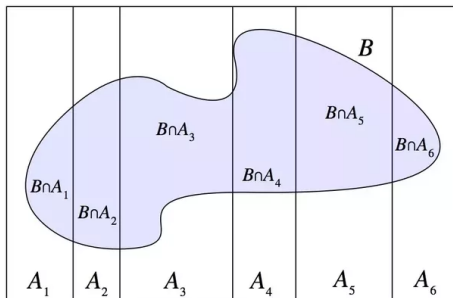


- The probability of wife is murdered
 $P(B) \approx 1,218/133,928,126 = 0.0000091$
 - “In 1999 in the United States, 1,218 women and 424 men were killed by an intimate partner.”

Law of Total Probability

- If events A_i are mutually exclusive and partition Ω , i.e., $A_i \cap A_j = \emptyset$ $\forall i, j$ and $\cup_{i=1}^n A_i = \Omega$, then

$$\begin{aligned} P(B) &= P(B|A_1) \cdot P(A_1) + P(B|A_2) \cdot P(A_2) + \cdots + P(B|A_n) \cdot P(A_n) \\ &= \sum_{i=1}^n P(B|A_i)P(A_i) \end{aligned}$$



Bayes' Theorem

- Law of Total Probability:

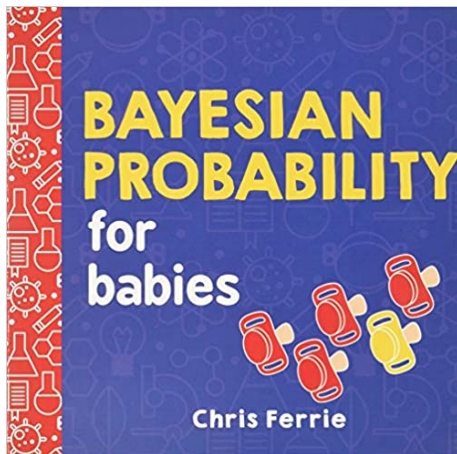
$$P(B) = \sum_{i=1}^n P(B|A_i)P(A_i)$$

- Let B be an event with non-zero probability, then the statement of Bayes' Theorem is:

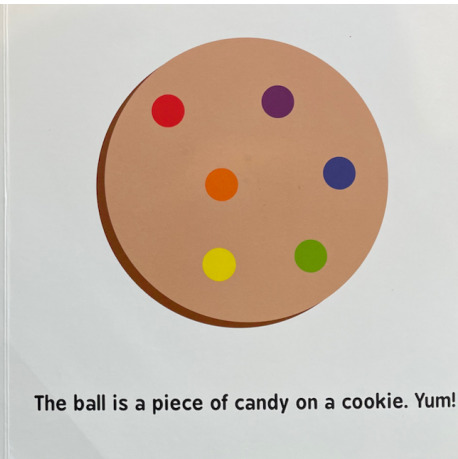
$$P(A_i|B) = \frac{P(B|A_i)P(A_i)}{P(B)} = \frac{P(B|A_i)P(A_i)}{\sum_{i=1}^n P(B|A_i)P(A_i)}$$

Examples

- How to explain Bayes' Theorem to a baby?



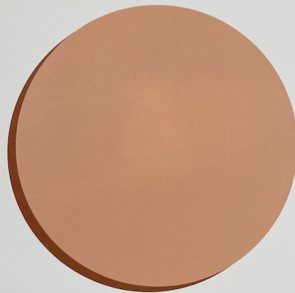
Bayesian Probability for Babies



Bayesian Probability for Babies



Some cookies have candy.



Some don't.

Bayesian Probability for Babies



Take a bite. It has no candy.



Did it come from a candy cookie?

Bayesian Probability for Babies



Either it came from a candy cookie,



or it didn't. What are the chances?

Bayesian Probability for Babies



If the cookie had no candy,
then every bite would have no candy.

$$\Pr(\text{no-candy bite} | \text{no-candy cookie}) = 1$$

The probability of a no-candy bite,
given a no-candy cookie, is 1.

Bayesian Probability for Babies



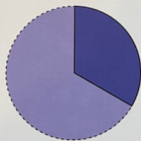
If the cookie had candy, then very few bites would have no candy.

$$\Pr(\text{no-candy bite} \mid \text{candy cookie}) = \frac{1}{3}$$

The probability of a no-candy bite, given a candy cookie, is 1/3.

Bayesian Probability for Babies

$$\Pr(\text{bite} | \text{solid}) > \Pr(\text{bite} | \text{candy})$$

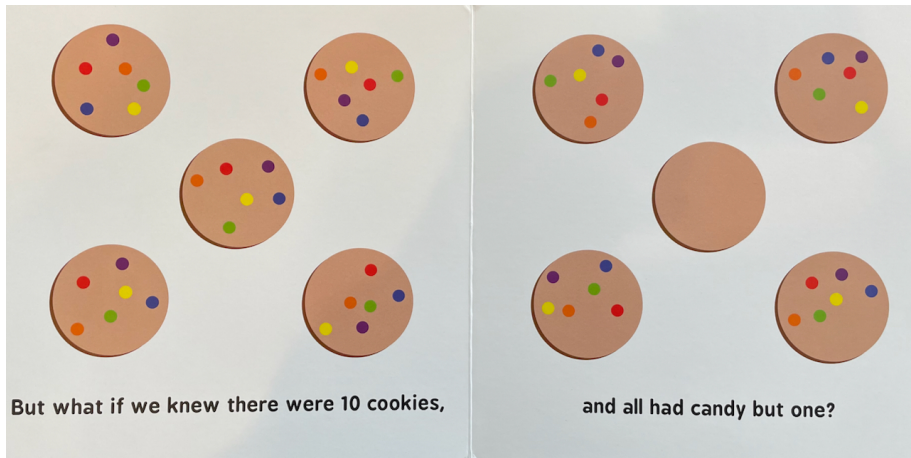


1 is greater than 1/3.

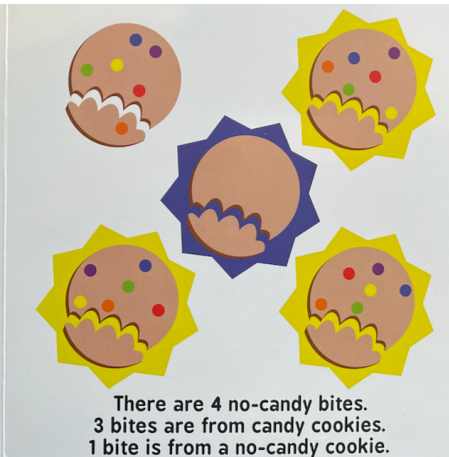


So the no-candy bite probably came from a no-candy cookie!

Bayesian Probability for Babies



Bayesian Probability for Babies



Bayesian Probability for Babies



1/3 of the candy cookies have a no-candy bite.



$$\Pr(\text{candy cookie} | \text{no-candy bite}) = \frac{3}{4}$$

The probability of a candy cookie with a no-candy bite is 3/4.

Bayesian Probability for Babies



This is the **prior distribution** of cookies.



This is the **posterior distribution** of cookies.

Bayesian Probability for Babies



This bite probably came from a candy cookie!

Now you know
**BAYESIAN
PROBABILITY!**

Bayesian Probability for Babies

- $P(A|B)$ vs. $P(B|A)$
 - The probability of having a no-candy bite when having a non-candy cookie
 - The probability of having a non-candy cookie when having a no-candy bite
- $P(A)$ vs. $P(A|B)$
 - The probability of having a non-candy cookie
 - The probability of having a non-candy cookie when having a no-candy bite
- Mathematics makes the problem easier to understand

In-class Practice: Bayesian Probability

- The probability of a no-candy bite is $1/3$, if it is a candy cookie.
Suppose there are nine candy cookies and one non-candy cookies in the box. If you have a no-candy bite, is it more likely from a candy cookie or a no-candy cookie?
- Formulation:
 - $B = 1/0$ indicates a candy/no-candy bite
 - $D = 1/0$ indicates a candy/no-candy draw
 - $P(B = 0|D = 0) = 1, P(B = 1|D = 0) = 1 - P(B = 0|D = 0) = 0;$
 $P(B = 0|D = 1) = 1/3,$
 $P(B = 1|D = 1) = 1 - P(B = 0|D = 1) = 2/3; P(D = 0) = 1/10,$
 $P(D = 1) = 1 - P(D = 0) = 9/10$
- Question: $P(D = 0|B = 0)?P(D = 1|B = 0)$, where $? \in \{<, =, >\}$

Solution: Bayesian Probability

- Formulation:

- $B = 1/0$ indicates a candy/no-candy bite
- $D = 1/0$ indicates a candy/no-candy draw
- $P(B = 0|D = 0) = 1$, $P(B = 1|D = 0) = 0$;
 $P(B = 0|D = 1) = 1/3$, $P(B = 1|D = 1) = 2/3$; $P(D = 0) = 1/10$,
 $P(D = 1) = 9/10$

- Solution:

$$\begin{aligned}P(D = 0|B = 0) &= \frac{P(B = 0|D = 0)P(D = 0)}{P(B = 0)} \\&= \frac{P(B = 0|D = 0)P(D = 0)}{P(B = 0|D = 0)P(D = 0) + P(B = 0|D = 1)P(D = 1)} \\&= \frac{1 \times 1/10}{1 \times 1/10 + 1/3 \times 9/10} \\&= 1/4\end{aligned}$$

Solution: Bayesian Probability

- Formulation:

- $P(B = 0|D = 0) = 1, P(B = 1|D = 0) = 0;$
 $P(B = 0|D = 1) = 1/3, P(B = 1|D = 1) = 2/3; P(D = 0) = 1/10,$
 $P(D = 1) = 9/10$

- Solution:

$$\begin{aligned}P(D = 1|B = 0) &= \frac{P(B = 0|D = 1)P(D = 1)}{P(B = 0)} \\&= \frac{P(B = 0|D = 1)P(D = 1)}{P(B = 0|D = 0)P(D = 0) + P(B = 0|D = 1)P(D = 1)} \\&= \frac{1/3 \times 9/10}{1 \times 1/10 + 1/3 \times 9/10} \\&= 3/4\end{aligned}$$

- Conclusion: $P(D = 0|B = 0) < P(D = 1|B = 0)$

Solution: Changing the Prior

- What if the numbers of candy and non-candy cookies are the same in the box
- Formulation:
 - $P(D = 0) = 1/2, P(D = 1) = 1/2$

- Solution:

$$P(D = 0|B = 0) = \frac{1 \times 1/2}{1 \times 1/2 + 1/3 \times 1/2} = 3/4$$

$$P(D = 1|B = 0) = \frac{1/3 \times 1/2}{1 \times 1/2 + 1/3 \times 1/2} = 1/4$$

- Conclusion: $P(D = 0|B = 0) > P(D = 1|B = 0)$

- Formulation:

- $D = 1/0$ indicates presence/absence of COVID-19
- $T = 1/0$ indicates positive/negative result
- $P(T = 1|D = 1) = 0.98$, $P(D = 1) = 1/14000$, and $P(T = 1|D = 0) = 0.01$
- What is $P(D = 1|T = 1)$?

- Solution:

$$\begin{aligned}P(D = 1|T = 1) &= \frac{P(T = 1|D = 1)P(D = 1)}{P(T = 1)} \\&= \frac{P(T = 1|D = 1)P(D = 1)}{P(T = 1|D = 0)P(D = 0) + P(T = 1|D = 1)P(D = 1)} \\&= \frac{0.98 \times 1/14000}{0.01 \times (1 - 1/14000) + 0.98 \times 1/14000} \\&\approx 0.007\end{aligned}$$

In-class Practice: COVID-19 in the U.S.

- Suppose the disease prevalence is about

$$P(D = 1) \approx \frac{1}{4.5}.$$

- Keep the same sensitivity and false-positive rate:

$$P(T = 1 \mid D = 1) = 0.98, \quad P(T = 1 \mid D = 0) = 0.01.$$

- What is $P(D = 1 \mid T = 1)$ now?

Solution: COVID-19 in the U.S.



$$P(D = 1 \mid T = 1) = \frac{0.98(1/4.5)}{0.01(1 - 1/4.5) + 0.98(1/4.5)} \approx \boxed{0.97}.$$

- The posterior probability changes dramatically because the disease prevalence changes.

- Populations are different
 - $P(D = 1) = 1/14000$ for China
 - $P(D = 1) = 1/4.5$ for U.S.
- Population is heterogeneous
 - $P(D = 1|G)$

Reagent test results	PCR Comparator		Subtotal
	positive	negative	
positive	105	3	108
negative	2	382	384
Subtotal	107	385	492

- True positive rate = $105/107 \approx 0.98$
- False positive rate = $3/385 \approx 0.01$

In-class Practice: COVID-19

- In China, suppose $P(D = 1) = \frac{1}{14000}$, $P(T = 1 \mid D = 1) = 0.98$, $P(T = 1 \mid D = 0) = 0.01$.
- What is

$$P(D = 1 \mid T = 1)?$$

Solution: COVID-19 in China

$$\begin{aligned} &P(D = 1 \mid T = 1) \\ &= \frac{P(T = 1 \mid D = 1)P(D = 1)}{P(T = 1 \mid D = 1)P(D = 1) + P(T = 1 \mid D = 0)P(D = 0)} \\ &= \frac{0.98(1/14000)}{0.98(1/14000) + 0.01(1 - 1/14000)} \\ &\approx \boxed{0.007}. \end{aligned}$$

- Even with a highly accurate test, a positive result does not necessarily imply that the patient has the disease.
- The low disease prevalence strongly affects the posterior probability.

In-class Practice: Basketball

A player makes a free throw with probability $p = 0.8$.

Suppose the player attempts 5 free throws and let X be the number of made free throws.

Find:

- the distribution of X ;
- $P(X = 4)$;
- $P(X \geq 4)$;
- $E(X)$.

Solution: Basketball

Let X be the number of made free throws. Then

$$X \sim \text{Binomial}(5, 0.8).$$

$$P(X = 4) = \binom{5}{4} (0.8)^4 (0.2) = \boxed{0.4096}.$$

$$P(X \geq 4) = P(X = 4) + P(X = 5) = 0.4096 + 0.32768 = \boxed{0.73728}.$$

$$E(X) = np = 5(0.8) = \boxed{4}.$$

In-class Practice: Baseball

A hitter gets a hit with probability $p = 0.30$ in each at-bat.

Suppose the player has 20 at-bats and let X be the number of hits.

Find

$$P(X \geq 8).$$

Then explain what happens to this probability if p increases from 0.30 to 0.40.

Solution: Baseball

$$X \sim \text{Binomial}(20, 0.30).$$

$$P(X \geq 8) = 1 - P(X \leq 7) = 1 - \text{pbinom}(7, 20, 0.30) \approx \boxed{0.2277}.$$

If p increases from 0.30 to 0.40, the distribution shifts toward larger values of X , so $P(X \geq 8)$ increases.

Sports Probability in R

For the basketball example,

$$X \sim \text{Binomial}(5, 0.8).$$

Calculate the probability of exactly four makes:

```
dbinom(4, size = 5, prob = 0.8)
```

Calculate the probability of at least four makes:

```
1 - pbinom(3, size = 5, prob = 0.8)
```

Generating Random Samples in R

- Suppose a black box where there are 5 red, 3 green, and 2 blue balls inside
- Generate the balls via `x <- c(rep("red", 5), rep("green", 3), rep("blue", 2))`
 - Draw a ball via `sample(x, size = 1)`
 - Draw three balls via `sample(x, size = 3)`
 - Draw three balls with replacement via `sample(x, size = 3, replace = TRUE)` or
`sample(c("red", "green", "blue"), size = 3, prob = c(0.5, 0.3, 0.2), replace = TRUE)`

The Function `sample()`

- Suppose x is the vector of elements being sampled
- Important arguments
 - `size`: The number of samples to generate
 - `prob`: A vector of probability weights for obtaining the elements of x
 - All elements should be positive
 - The sum of all elements should equal to 1
 - `replace`: a logical value that indicates if sampling the elements of x with replacement

Generating Random Samples in R

- Mimic the experiment of flipping a coin 10 times
 - `sample(c("T", "H"), size = 10, replace = TRUE)`
 - `sample(c(0, 1), size = 10, replace = TRUE)`
 - `sample(c(0, 1), size = 10, replace = TRUE, prob = c(0.1, 0.9))`

Summary of Probabilities

- Probability $(\Omega, \mathcal{F}, P)$
 - Sample space Ω : Each of the K elements is an outcome
 - Set of events $\mathcal{F}$: Each of the 2^K elements is a question that we can ask from the experiment
 - Probability measure P : A function from $\mathcal{F}$ to $[0, 1]$
- Conditional probability: $P(A|B)$
 - Independence: $P(A \cap B) = P(A)P(B)$ or $P(A|B) = P(A)$
 - In general $P(A|B) \neq P(B|A)$
 - Bayes' theorem:

$$P(A|B) = \frac{P(B|A)P(A)}{\sum_{i=1}^n P(B|A_i)P(A_i)}$$

Summary of Bayesian Probabilities

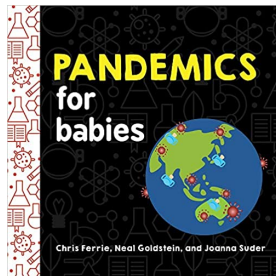
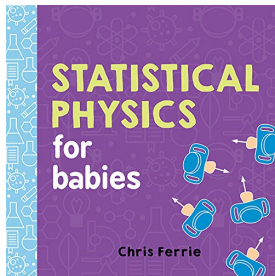
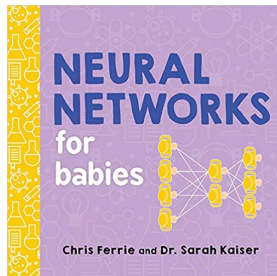
- Let $A = y$ be the observed data and $B = \theta$ be the parameter

$$P(\theta|y) = \frac{P(\theta)P(y|\theta)}{P(y)}$$

- Conditional probability: $P(\theta|y)$
- Marginal probability: $P(y)$
 - Law of total probability
 - Discrete parameter: $P(y) = \sum_{k=1}^K P(y|\theta_k)P(\theta_k)$
 - Continuous parameter $P(y) = \int P(y|\theta)dP(\theta)$
- Joint probability: $P(y, \theta)$
 - Chain rule: $P(y, \theta) = P(y|\theta)P(\theta)P(y, \theta_1, \theta_2) =$
 $P(y|\theta_1, \theta_2)P(\theta_1|\theta_2)P(\theta_2)P(y, \theta_1, \theta_2, \theta_3) =$
 $P(y|\theta_1, \theta_2, \theta_3)P(\theta_1|\theta_2, \theta_3)P(\theta_2|\theta_3)P(\theta_3) \quad \vdots$

Bayesian Probability for Babies

- Other statistics related book for babies



After-class Reading

- *A First Course in Bayesian Statistical Methods* by Peter Hoff
 - Chapter 2 Belief, probability and exchangeability
 - Section 2.1 Belief functions and probabilities
 - Section 2.2 Events, partitions and Bayes' rule
 - Section 2.3 Independence